

Title: Smart Recruitment Assistant System

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Problem Statement

Core Challenge: Manual resume screening is slow, biased, and costly.

- AI Solution: Predict candidate behavior and automate initial screening.

Data Preprocessing & Engineering

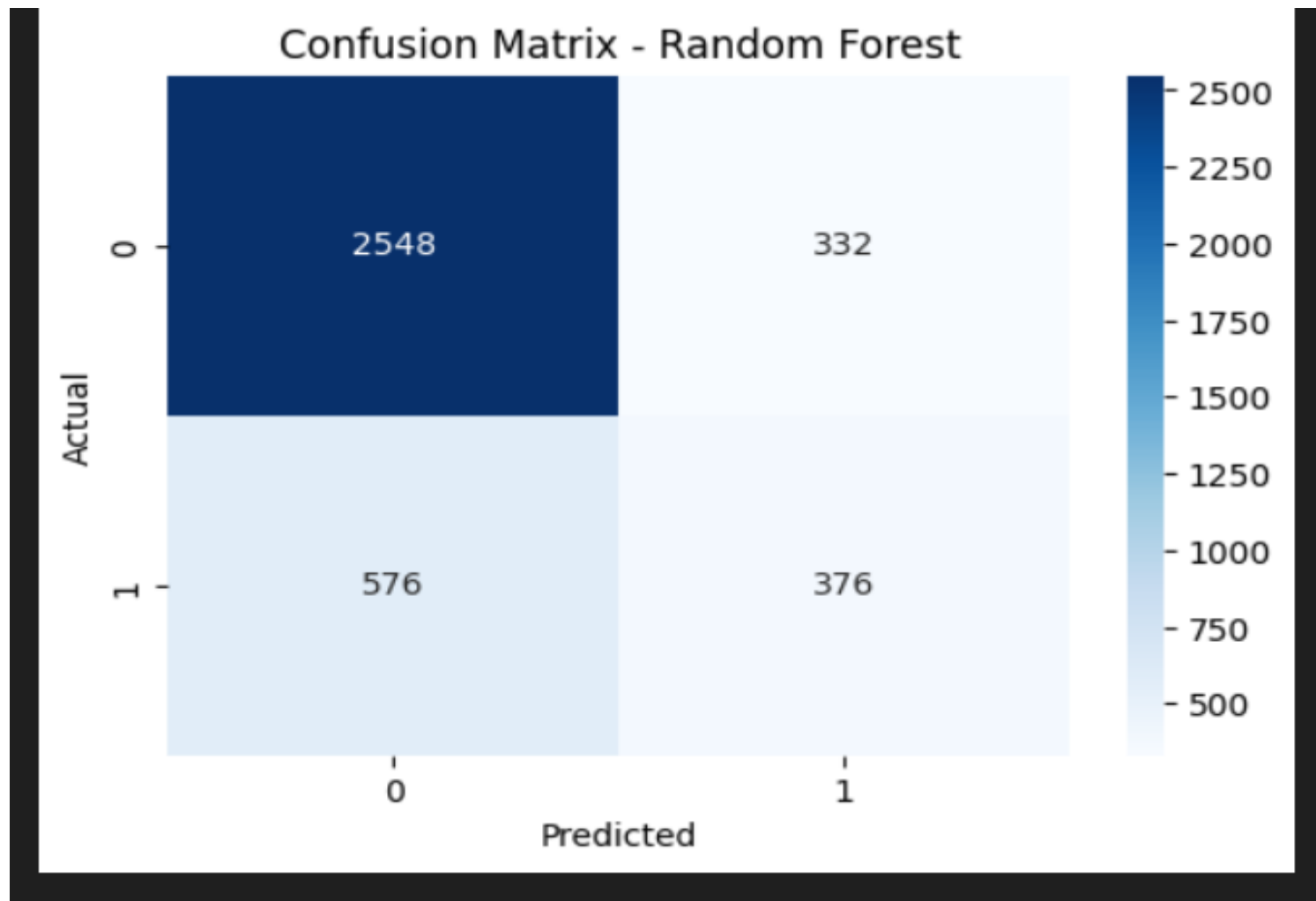
- Total Profiles: 19,158 records.
- Actions taken: Handled missing data + Applied One-Hot Encoding.
- Split: 80% Training, 20% Testing.

Machine Learning Modeling

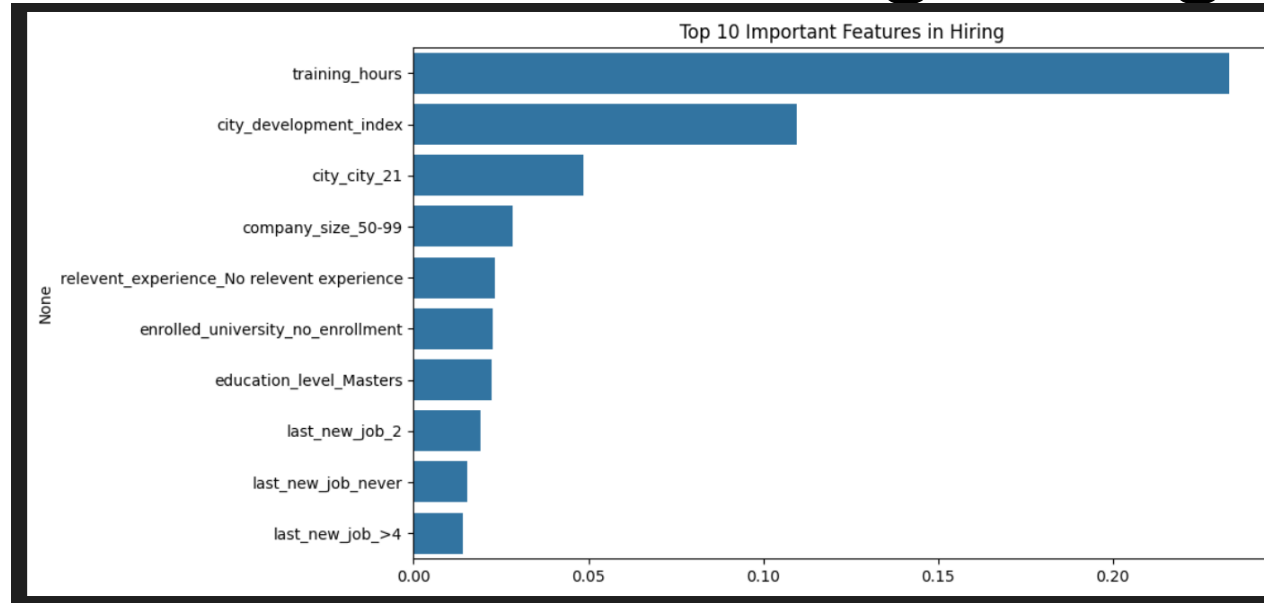
- Model 1: Logistic Regression (Baseline).
- Model 2: Random Forest Classifier (Advanced).

Results & Evaluation

- Comparison: Random Forest achieved higher accuracy.



Bonus: Smart Ranking & Insights



- **Candidate 2878** — Hiring Probability: **99%** (0.99)
- **Candidate 3111** — Hiring Probability: **98%** (0.98)
- **Candidate 6011** — Hiring Probability: **95%** (0.95)
- **Candidate 1277** — Hiring Probability: **95%** (0.95)
- **Candidate 5157** — Hiring Probability: **94%** (0.94)
- **Candidate 2701** — Hiring Probability: **94%** (0.94)
- **Candidate 5478** — Hiring Probability: **94%** (0.94)
- **Candidate 1981** — Hiring Probability: **94%** (0.94)

Business Impact & Value

- Reduces HR screening time by up to 70%.
- Data-driven, unbiased decision-making process for enterprise hiring.